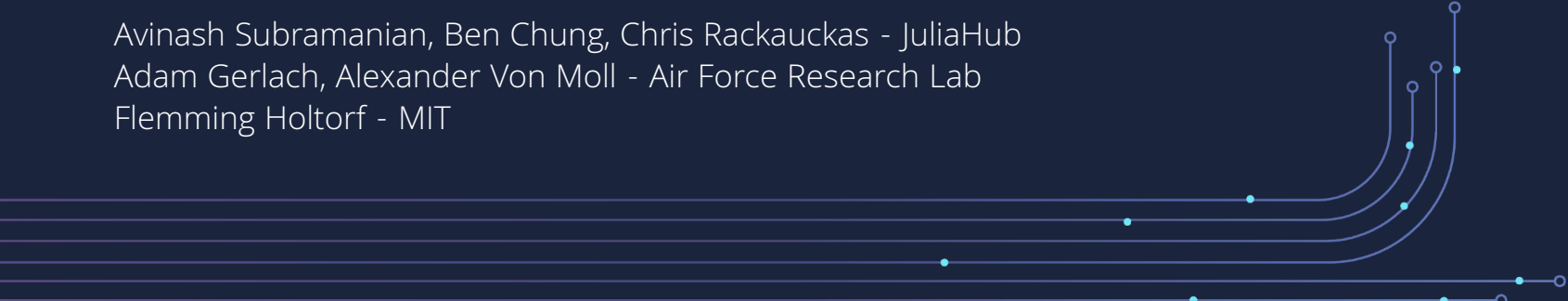


Optimal Uncertainty Quantification of SciML models

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July, 2025

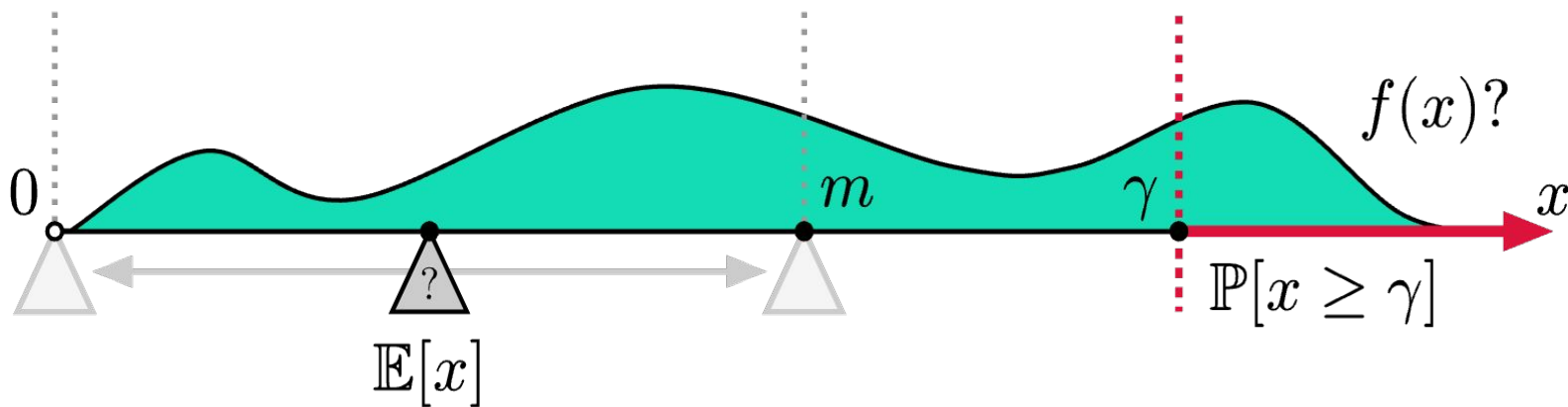
A decorative graphic on the right side of the slide, consisting of several horizontal lines of varying lengths and colors (light blue, teal, and white). These lines curve upwards and to the right, ending in small circles of the same colors. The overall effect is a stylized, modern design element.

Illustrative Problem

Imagine you have a distribution of sand on a see-saw:

You know a see-saw is balanced, so the forces are the same on both sides

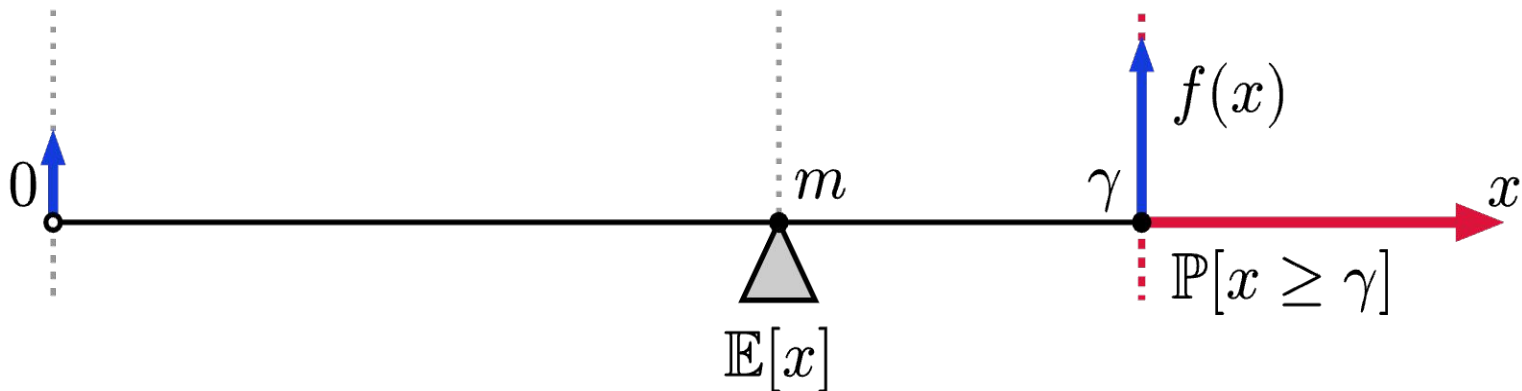
What is the maximum percentage of the sand that is to the right of the point γ ?



To figure out the maximum, we can construct a worst-case scenario:

To get the most sand on the right, we want to push down on the left as hard as possible, so move all of the sand as far from the center as you can. And to get less of a push on the right, move all of the sand as close to the center as possible.

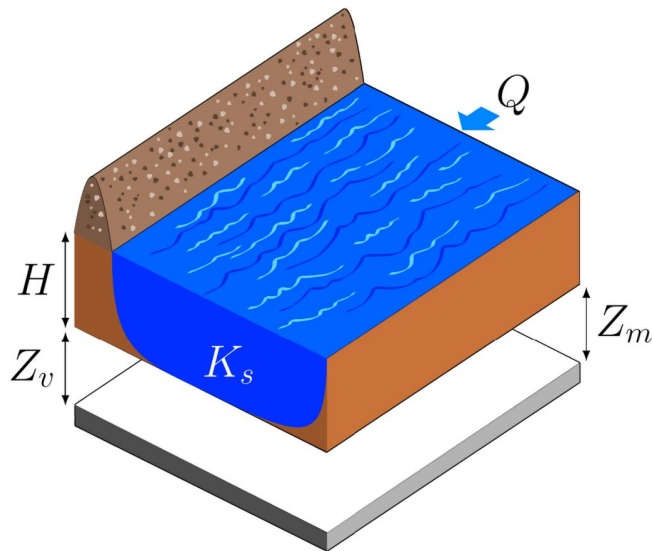
With sand only in those two spots, how much sand can you put on the right and still balance the see-saw?



Why does this matter? The Certification Problem

- Determine probability of failure given uncertainties in parameters

Example: Build a flood barrier that is safe with probability 99%



River height: $H = f(Q, K_s, Z_v, Z_m)$

Solve for the worst-case probability:

$$\sup_{\mu \in \mathcal{A}} \mathbb{P}[H \geq h]$$

CERTIFY: If objective value is less than 1%
Else: **DECERTIFY**

Distributionally-Robust Uncertainty Quantification

❖ Formulate an **Optimization Problem**

Objective: $\sup_{\mu \in \mathcal{A}} \mathbb{P}[H \geq h]$

Constraints: All known information. **Moment information** and **Bounds**

Search: over space of feasible probability distributions for **worst-case distribution**

Solution: Leverage **Reduction Theorems**

Option 1. **Extreme Points** of Measure Sets (Winkler)

Option 2. **Canonical Moments** (Stenger)

Key idea:
**Worst-case
distribution is
discrete**

Winkler, G. 1988. Extreme Points of Moment Sets. Mathematics of Operations Research

Stenger J., Gamboa F., Keller M., Iooss B, Optimal Uncertainty Quantification of a Risk Measurement from a Thermal-Hydraulic Code using Canonical Moments

QN: Why not Monte Carlo Simulation?

- Assumes certain **prior** distributions on uncertain parameters (e.g., Uniform, Normal etc.)

Takes a large number of samples, push forward through the model, **estimate posterior** probabilities of observable of interest

- Issue 1: We don't know the full prior distribution.

“Your prior can bite you on the posterior”

- Issue 2: Not suitable for **safety-critical systems**
- Issue 3: Monte Carlo simulation may require a huge number of samples

Stark, P. "Your prior can bite you on the posterior: contrasting Bayesian and frequentist measures of uncertainty." In JPL Science Visitor and Colloquium Program-Earth Science Seminar, Sept. 1, 2020.

Let's use OptimalUncertaintyQuantification.jl!

1. Specify random variables & Information constraints

```
rand_vars = @random_variables begin
    Independent(Q, bounds = (160.0, 3580.0))
end

constraints = [
     $\mathbb{E}(Q) \sim 1320.42$ 
]

admissible_set = AdmissibleSet(rand_vars, constraints)
```

2. Specify probability objective & reduction theorem

```
H = (Q/(300*K_s*((Z_m - Z_v)/5000)^(0.5)))^(3/5)
objective=  $\mathbb{P}(H \geq h)$ 
ouq_sys = OUQSystem(; objective,
                    admissible_set,
                    StengerCanonicalMoments())
```

This generates the following optimization problem:

```
Min w_Q_1*on1 + w_Q_2*on2
Subject to
  in_con_Q1_on - (2.0 + (-0.034768162717021094 * (x_Q_1 ^ 0.6))) = 0
  in_con_Q1_off - (-2.00000000001 + (0.034768162717021094 * (x_Q_1 ^ 0.6))) = 0
  in_con_Q2_on - (2.0 + (-0.034768162717021094 * (x_Q_2 ^ 0.6))) = 0
  in_con_Q2_off - (-2.00000000001 + (0.034768162717021094 * (x_Q_2 ^ 0.6))) = 0
  w_Q_1 + w_Q_2 = 1
  w_Q_1*x_Q_1 + w_Q_2*x_Q_2 = 1320.42
  on1 --> {in_con_Q1_on ≤ 0}
  on2 --> {in_con_Q2_on ≤ 0}
  !on1 --> {in_con_Q1_off ≤ 0}
  !on2 --> {in_con_Q2_off ≤ 0}
  w_Q_1 ≥ 0
  w_Q_2 ≥ 0
  x_Q_1 ≥ 160
  x_Q_2 ≥ 160
  w_Q_1 ≤ 1
  w_Q_2 ≤ 1
  x_Q_1 ≤ 3580
  x_Q_2 ≤ 3580
  on1 binary
  on2 binary
```

Results - Flood Problem:

Barrier Height:

Worst-case probability (%)

2.0 meters

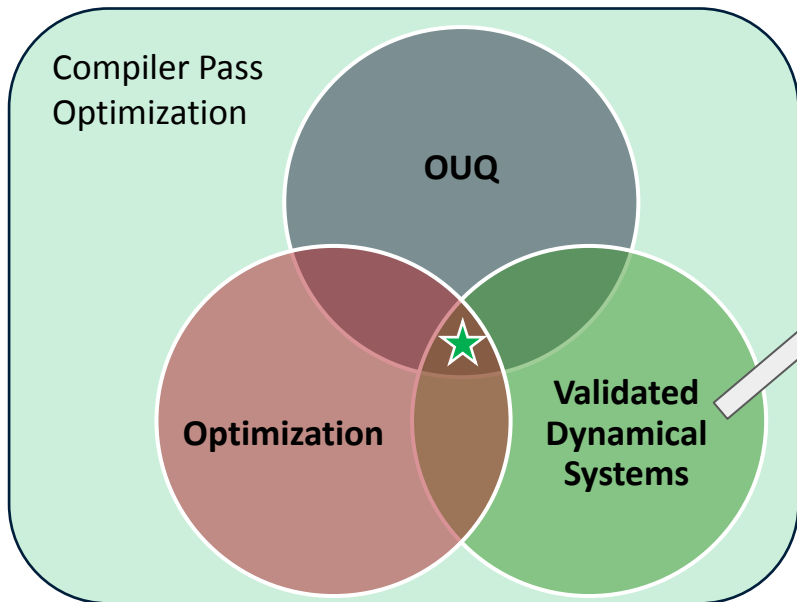
17.0% => **DECERTIFIED**

4.6 meters

0.0% => **CERTIFIED**

For any uncertainty distribution with the chosen means and variances, a barrier height of 4.6 is safe!

OUQ of Dynamical Systems



Leverage **JuliaReach** + **JuliaIntervals** tools:

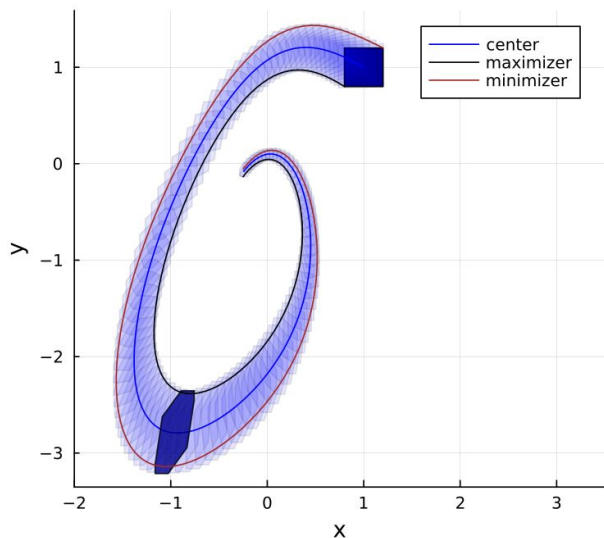
- Bound the ODE solution
- [ReachabilityAnalysis.jl](#) + TaylorModels.jl

Do the airplanes collide? Validated ODE + Optimization + OUQ

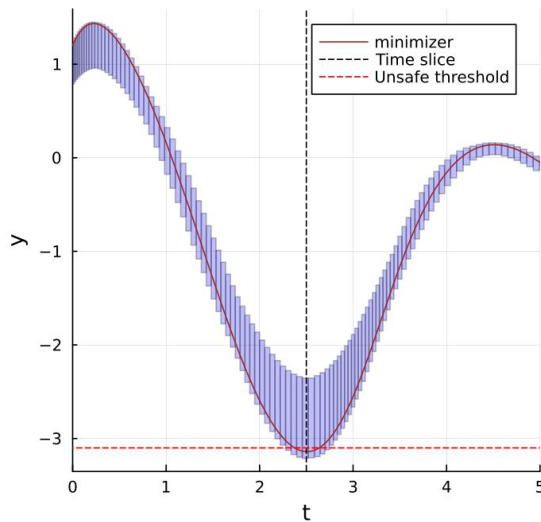
Moore-Greitzer Model:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} -y - \frac{3}{2}x^2 - \frac{1}{2}x^3 - \frac{1}{2} \\ 3x - y \end{bmatrix}$$

Initial conditions are uncertain: $(x_0, y_0) \in [0.8, 1.2]^2$



(a) Reachable set in phase space



(b) Reachable set, unsafe threshold & trajectory of minimizer

Find **worst-case probability** that state **y** is below unsafe threshold at $t = 2.5$ s

Results:

Design 1:

$$(x_0, y_0) \in [0.8, 1.2]^2 \subseteq \mathbb{R}^2$$

$$\mathbb{E} [x_0 \mid \mu_{x_0}] = 1.0,$$

$$\mathbb{E} [y_0 \mid \mu_{y_0}] = 1.0$$

Worst-case probability (%)

31.3% => **DECERTIFIED**

Design 2: Same bounds but have higher order moments

$$(x_0, y_0) \in [0.8, 1.2]^2 \subseteq \mathbb{R}^2$$

$$\mathbb{E} [x_0 \mid \mu_{x_0}] = 1.0,$$

$$\mathbb{E} [y_0 \mid \mu_{y_0}] = 1.0$$

$$\mathbb{E} [x_0^2 \mid \mu_{x_0}] = 1.0,$$

$$\mathbb{E} [y_0^2 \mid \mu_{y_0}] = 1.0$$

0.0% => **CERTIFIED**

Note: Only 1 feasible trajectory

Conclusions and Future Work

- OUQ provides a rigorous iterative workflow for **certification of safety-critical systems**
- Distributionally robust - does not require unjustified assumptions on priors
- Computationally tractable -> Leverages **reduction theorems**

Future Work:

1. Guaranteed global optimization: [SpatialBranchAndBound.jl](#)
2. Parametric sensitivity analysis

This paper is based on work performed at the Air Force Research Laboratory (AFRL) Control Science Center. DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited. AFRL-2025-2730; Cleared 05/29/2025.

Results:

Design 1:

$$(x_0, y_0) \in [0.8, 1.2]^2 \subseteq \mathbb{R}^2$$

$$\mathbb{E} [x_0 \mid \mu_{x_0}] = 1.0,$$

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Worst-case probability (%)

31.3% => DECERTIFIED

Results:

Design 1:

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$$\mathbb{E}[y_0 \mid \mu_{y_0}] = 1.0$$

Worst-case probability (%)

31.3% => **DECERTIFIED**

Design 2: Assume **tighter bounds** on initial states (e.g., with more experiments)

$$(x_0, y_0) \in [0.82, 1.18]^2 \subseteq \mathbb{R}^2,$$

$$\mathbb{E}[x_0 \mid \mu_{x_0}] = 1.0,$$

$$\mathbb{E}[y_0 \mid \mu_{y_0}] = 1.0$$

26.5% => **DECERTIFIED**

Results:

Design 1:

$$(x_0, y_0) \in [0.8, 1.2]^2 \subseteq \mathbb{R}^2$$

$$\mathbb{E} [x_0 \mid \mu_{x_0}] = 1.0,$$

$$\mathbb{E} [y_0 \mid \mu_{y_0}] = 1.0$$

Worst-case probability (%)

31.3% => **DECERTIFIED**

Design 3: Even **tighter bounds** on initial states (e.g., with more experiments)

$$(x_0, y_0) \in [0.84, 1.16]^2 \subseteq \mathbb{R}^2,$$

$$\mathbb{E} [x_0 \mid \mu_{x_0}] = 1.0,$$

$$\mathbb{E} [y_0 \mid \mu_{y_0}] = 1.0$$

0.0% => **CERTIFIED**

Distributionally-Robust Uncertainty Quantification

❖ Formulate an **Optimization Problem**

Objective: $\sup_{\mu \in \mathcal{A}} \mathbb{P}[H \geq h]$

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Owhadi, H., Scovel, C., Sullivan, T.J., McKerns, M. and Ortiz, M., 2013. Optimal Uncertainty Quantification. SIAM Review, 55(2), pp.271-345.

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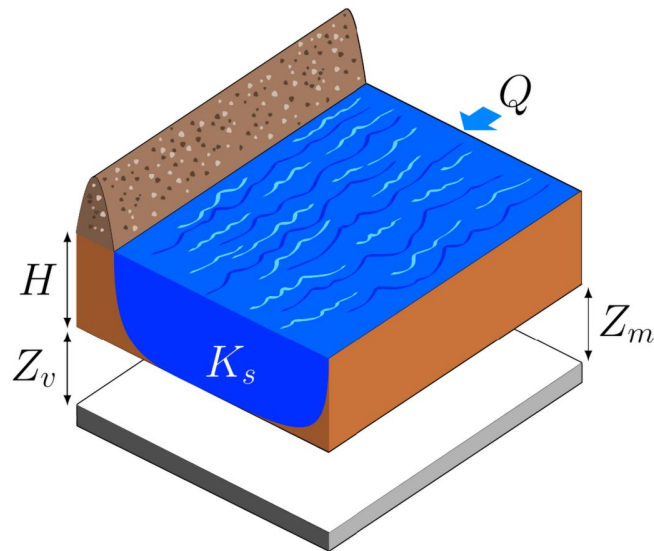
ISSUE:
Search over an
Infinite-dimensional measure space!

Owhadi, H., Scovel, C., Sullivan, T.J., McKerns, M. and Ortiz, M., 2013. Optimal Uncertainty Quantification. SIAM Review, 55(2), pp.271-345.

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Mathematical Version: The worst-case distribution is discrete

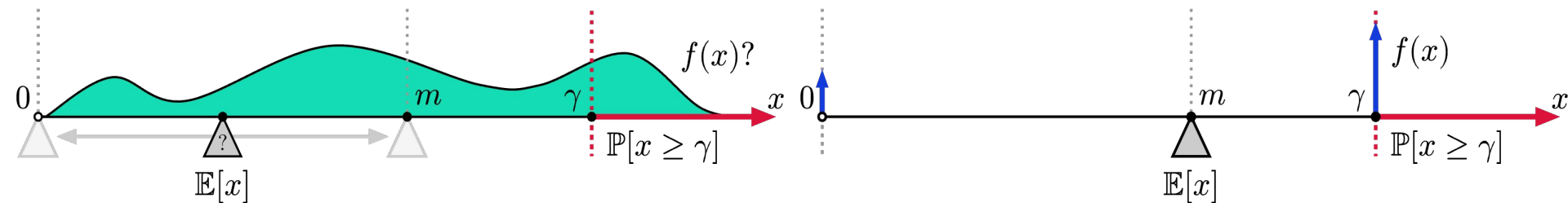
Extreme Measure Theorem:

Measure that extremizes expectation/probability is discrete and supported on at most $N+1$ points

$$\sup_{\mu \in \mathcal{A}} \mathbb{P}[x \geq \gamma \mid \mu]$$

$$\mathbb{E}[x] \leq m, \text{ for } 0 < m \leq \gamma$$

Key: we can calculate a bound that works for every possible distribution!



Winkler, G. 1988. Extreme Points of Moment Sets. Mathematics of Operations Research